


Designed by Remus Teodorescu 2026

LECTURE 11.1

# Vision-Based Navigation

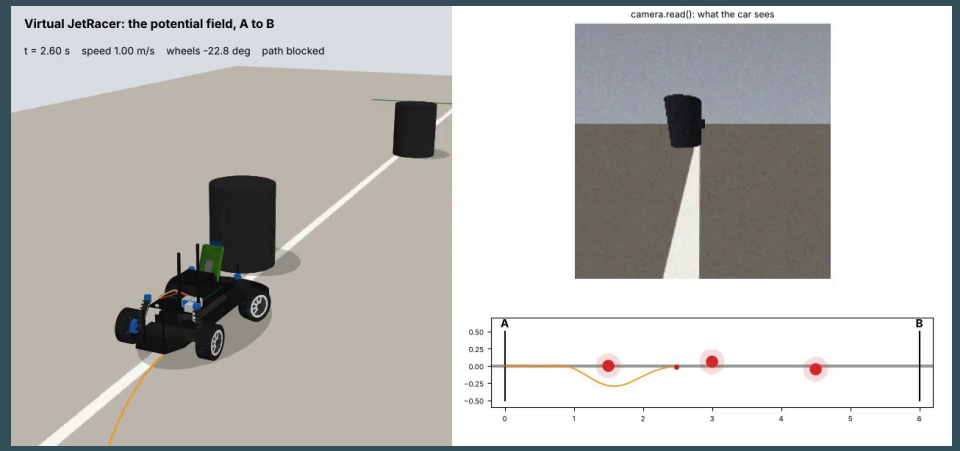

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Deep Learning for Engineering · 2026

NEED · RULES · NETWORKS




*Cars, warehouse robots, underwater vehicles and humanoid robots steer by what a camera sees, answering two questions: where is the path, and how do I steer onto it? The lecture answers both with fixed rules, then with a convolutional network and the GPU that makes it fast enough.*

# Learning Objectives

## SECTIONS, IN ORDER

1. **The need** — vehicles and robots that steer by camera
2. **Conventional vision** — fixed rules, and where they fail
3. **Deep learning** — a network that steers, its data, its GPU

*Two questions run through the lecture: where is the path, and how do I steer onto it? Each is answered first in words, then in mathematics - by fixed rules, then by a network - on Ex_11's virtual JetRacer, where the true position of the line is known.*

## READ, AND WHAT EACH ANSWERS

| | |
|---|---|
| **UDL ch. 1** | where is deep learning applied? |
| **UDL ch. 10** | why do images need convolutions? |
| **UDL ch. 8** | what must the training data show? |

# Recap of L10.2 Electrolysers

## 1 – 2 · A NETWORK FOR THE FUEL CHANNEL

**1** **How can one network learn the steam fraction, temperature and current density along the channel at every operating point?**

The network takes the position along the channel, the cell voltage and the inlet temperature, and returns the three fields. It is trained over 1.15 to 1.35 volts and 750 to 850 degrees until the three laws hold at points over the channel and the operating range; the inlet values are built into the form of the solution, and the current is kept below the limiting current. Any operating point is then read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

**2** **Why does the local current make the channel harder to solve, and how is the network's answer checked?**

The current has no equation of its own along the channel: at every point it is the root of the charge transfer at the local steam fraction and temperature, so no field can be solved first. The network takes it as a third residual, in units of 50 millivolts so it weighs like the balances. Scored against a boundary-value solver at twelve operating points, it is within 0.0022 of the steam fraction and 0.19 kelvin, three times closer than finite volumes on 50 cells. *Ex_10.2 sections 3 – 5*

## 3 – 4 · THE OPTIMAL VOLTAGE

**3** **How can the optimal cell voltage be found with a network, and what does the gradient save?**

The cost of an hour, the electricity bought less the hydrogen sold plus the share of the stack aged, is read from the network's fields, with the 10 kelvin limit as a penalty. Autograd through the network gives its derivative in the voltage, and gradient descent finds 1.250 volts in seconds; without the limit, 1.167 volts, where the cell cools 25 kelvin. A sweep of 81 solver runs agrees. The prices and the ageing law are assumptions, so the optimum is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

**4** **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of voltage and temperature: what produces more now ages the cell faster. A solver gives the cost, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives every gradient by automatic differentiation. The degradation constants here are assumed, and published rates span an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

# Autonomous Vehicles and Robots

**Vision-based navigation** — steering a vehicle or a robot from what its camera sees: where the free path is, where it leads, and what is in the way. The camera is cheap and carries the most information of any sensor, but turning its pixels into a decision is the hard part.

## FOUR OF THEM, AND THE LOOP THEY SHARE



*robotaxi*




*warehouse AGV*



*humanoid*



*underwater vehicle*

**see** one frame → **decide** path, obstacles → **act** steer, speed → **in time** fast enough

*Photographs: Wikimedia Commons. This lecture is the path; L11.2 is what blocks it.*

## WHAT MAKES THE CAMERA'S JOB HARD

| | specific challenges | beside the camera |
|---|---|---|
| road car | night, rain, low sun; 3.6 m blind per 0.1 s | radar, LiDAR, GNSS |
| robotaxi (Cybercab) | no wheel or pedals: no driver to take over | none: cameras only |
| warehouse AGV | shiny floors, narrow aisles, people and forklifts | safety LiDAR, floor markers |
| underwater vehicle | light gone within metres; murky water | sonar, Doppler log, inertial |
| humanoid robot | camera sways; stairs need depth | depth camera, joint angles |

# The JetRacer Pro

**JetRacer Pro** — a small car that computes on board: a Jetson Nano runs the network on its GPU, a wide camera is its only sensor, a servo steers and a brushed motor drives. It is the scaled-down case of the machines on the previous slide, with the same loop: see, decide, act.

## WHERE A ROW OF PIXELS LANDS ON THE FLOOR

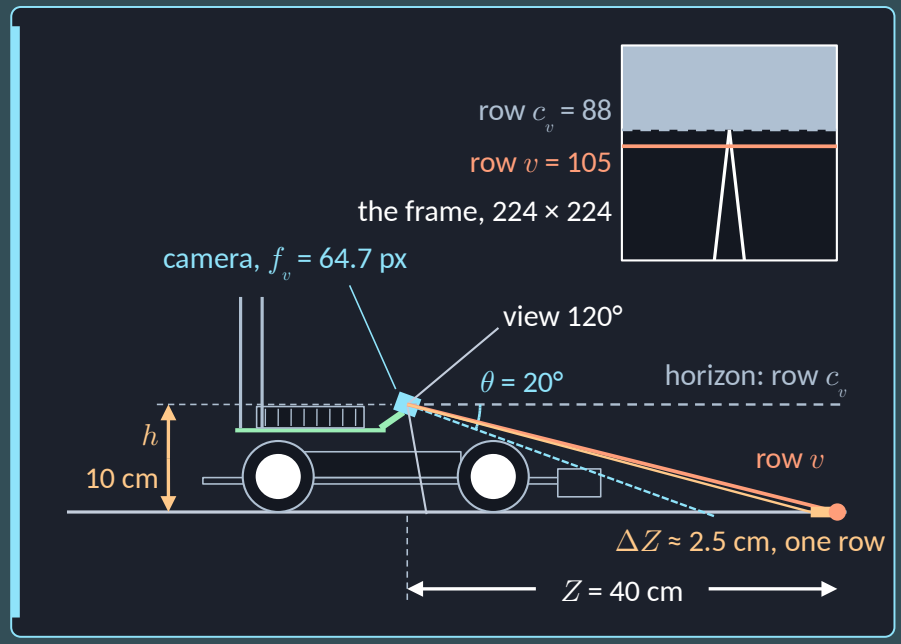
$$v - c_v = \frac{f_v\, h}{Z}, \qquad \Delta Z \approx \frac{Z^2}{f_v\, h}$$

$v$ the image row, $c_v$ the horizon's row, $f_v$ the focal length in pixels
$h$ the camera's height, $Z$ the distance ahead, $\Delta Z$ the floor one row covers

## THE VERIFIED SPECIFICATION

| compute | Jetson Nano 4 GB, 128-core Maxwell GPU |
|---|---|
| camera | IMX219-160: 8 MP, 160° lens; the network reads 224 × 224 |
| steering | Ackermann, 6 kg·cm servo; 4WD, RC380 brushed motor |
| pack | 8.4 V, four 18650 cells, 2S2P; no encoder |

## THE CAMERA ON THE CAR, SIDE VIEW




*To scale. The camera's height, tilt and field of view are the virtual JetRacer's, assumed; they are measured on the car.*

# Vision by Hand-Written Rules

**Conventional vision** — fixed steps chosen by an engineer. The white tape reflects more light than the dark floor, so pixels brighter than the floor mark the line; the centre of the bright run is where the tape crosses the row - the point to steer toward. Nothing is learned from data.

## THE DETECTOR, AS EX_11 WRITES IT

$$w_j = \max\left(I_j - \bar{I} - 0.15,\ 0\right), \qquad \hat{x} = \frac{\sum_j w_j x_j}{\sum_j w_j}$$

$I_j$ the brightness of pixel $j$, $\bar{I}$ the floor's median, $x_j$ its column
$w_j$ its weight, $\hat{x}$ the estimated column of the line

### FROM A FRAME TO A COMMAND

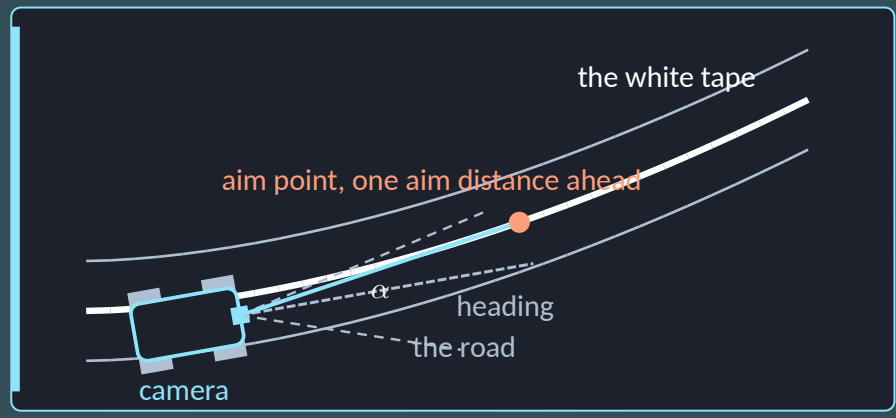
**frame** grey pixels → **threshold** set by hand → **line** bright centre → **rule** steer to it

every step has a number chosen by an engineer, for one floor and one light

## THE STEERING RULE

$$\delta = \arctan\frac{2L\sin\alpha}{\ell_d}$$

$\delta$ the steering angle, $\alpha$ the angle to the aim point
$L$ the wheelbase, 0.20 m, $\ell_d$ the aim distance, 0.40 m




*The rule is pure pursuit: steer onto the circle through the aim point; $L$ and $\ell_d$ assumed until measured.*

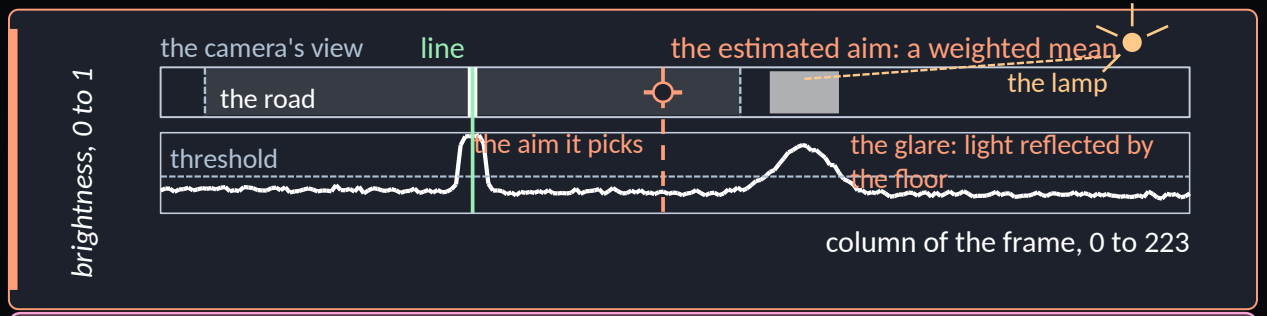
# Where Conventional Vision Fails

**Robustness** — keeping the error within what was agreed when the scene changes: the light, a reflection, a shadow, a new floor, an unseen object. A hand-tuned rule is robust only to the changes its designer thought of.

## WHY THE ESTIMATED AIM MOVES TOWARD THE GLARE

$$\hat{x} = \frac{W_\ell x_\ell + W_g x_g}{W_\ell + W_g}$$

$W_\ell, x_\ell$, $W_g, x_g$ the summed weight and mean column of the bright pixels at the tape and at the glare
$\ell$ and $g$ are our labels, for the analysis : the rule reads only brightness and sums every bright pixel




**TALK · 2 MIN** How would you explain, in your own words, why a rule that works on one floor can fail on another? Then, in the plot above, why does the estimate land between the line and the glare?

## THE DETECTOR ON 300 FRAMES

| the scene | mean error, fraction of frame width |
|---|---|
| along the line | 0.007 |
| beside the line | 0.021 |
| with glare | 0.083 |
| the agreed error | 0.03 |

*The formula is a weighted mean: each bright column counts in proportion to its brightness above the floor. It should return the tape's column, the aim - but the glare's 57 % of the weight moves it to column 109, off the tape's 68. Ex_11 section 3 draws this frame.*

# Steering From a Frame

**Vision policy** — a convolutional network that maps one camera frame to one action. It learns its own features from frames whose true answer is known, instead of following rules written by hand. One output between −1 and +1 makes a steerer; two outputs, free or blocked, detect an obstacle.

## ONE CONVOLUTION, AS IN L5.1

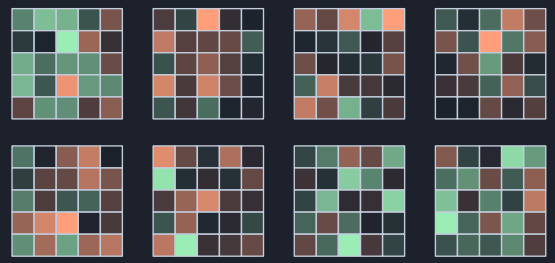
$$y_{ij} = \sigma\Big(\sum_{m,n} w_{mn}\, x_{i+m,\, j+n} + b\Big)$$

$x$ the frame, $w$ the kernel, $b$ its bias, $\sigma$ the ReLU, $y$ the feature map

### THE SAME SUM, TWO WAYS TO CHOOSE THE KERNEL



set by hand
Sobel: an edge



learned from the frames: the eight kernels of the first layer

### MEAN ERROR ON 300 HELD-OUT FRAMES

| | detector | network |
|---|---|---|
| along the line | 0.007 | 0.018 |
| beside the line | 0.021 | 0.035 |
| with glare | 0.083 | 0.055 |
| pixels | 112, grey | 224, colour |
| training | none | 80 s on a CPU |

*Four convolutions and two dense layers, trained on the squared error to the true position. Shown glare in training, the network is the better of the two with glare; the price is training, and about twice the time per frame on a CPU.*

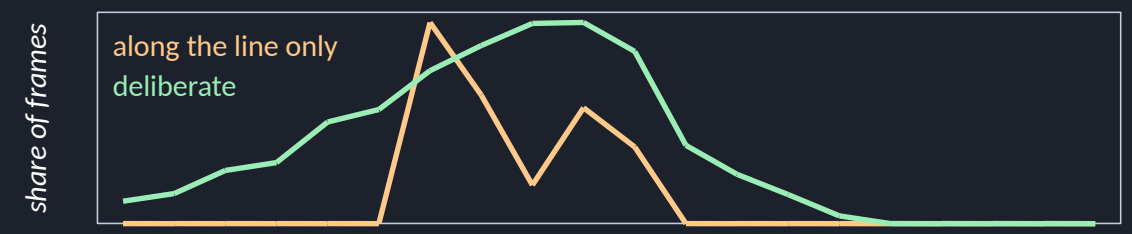
# Collect the Data Deliberately

**Deliberate collection** — choosing the frames the policy will be trained on. The network learns only what the pictures show: the floor, the mount and the light it will drive in, and on purpose the positions the car reaches when it is already going wrong, labelled with the way back.

## THE LOSS SEES ONLY THE TRAINING FRAMES

$$\mathcal{L} = \mathbb{E}_{x \sim p_{\text{train}}}[(\hat{y} - y)^2]$$

$\mathcal{L}$ the loss, $p_{\text{train}}$ where the line is in the training frames; the car meets $p_{\text{drive}}$




A policy shown only perfect driving has never seen a mistake, so it cannot correct one: small errors add up, lap after lap.

## THE SAME NETWORK, TRAINED TWO WAYS

| mean error | along only | deliberate |
|---|---|---|
| along the line | 0.006 | 0.018 |
| beside the line | 0.185 | 0.035 |
| with glare | 0.186 | 0.055 |

*Trained only along the line the network is 5 times worse exactly where a drifting car needs it. Same network, same number of frames, same training time: only the collection differs.*

# Why a GPU

**GPU** — a processor with many small cores that run the same operation on different data at the same time. Every output of a convolution is its own sum of products over a small window, independent of the others, so a GPU computes them all at once.

## THE DISTANCE DRIVEN WITHOUT NEW INFORMATION

$$d = \frac{v}{f} = v\,(t_{\text{cap}} + t_{\text{pre}} + t_{\text{net}} + t_{\text{act}})$$

$d$ the blind distance, $v$ the speed, $f$ decisions per second
$t$ the times to capture, preprocess, run the network and act

### ONE CONVOLUTION, ALL OUTPUTS TOGETHER

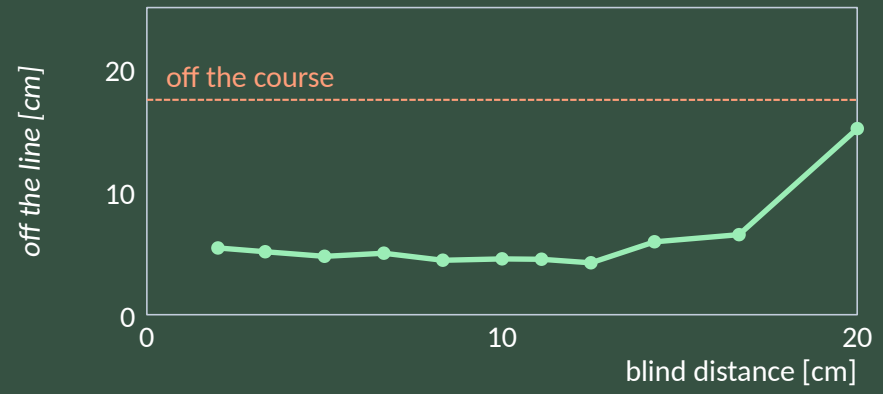


one core per output

the frame, one 3 × 3 window every output at once

**TALK · 2 MIN** How would you explain, in your own words, why a slightly more accurate but slower network can be a worse driver? Then, in the plot on the right, where does the car start to swing wide?

## THE LAP, AND THE WORK PER FRAME




$$\text{MAC} = H_{\text{out}}\, W_{\text{out}}\, C_{\text{out}}\, C_{\text{in}}\, k^2$$

$H_{\text{out}}, W_{\text{out}}$ the output size, $C_{\text{in}}, C_{\text{out}}$ the channels, $k$ the kernel

| | |
|---|---|
| **Ex_11's network** | 17 million a frame |
| **ResNet-18** | 1.8 billion, at 224 × 224 |
| **Nano GPU** | 472 GFLOPS at its peak |

# Advanced Topics

## RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic | what it adds | read | where it makes a difference |
|---|---|---|---|
| semantic segmentation | which pixels are free to drive on | UDL ch. 10 | floors with no line |
| depth from one camera | distance from a single frame | UDL ch. 10 | no range sensor on board |
| vision transformers | attention over patches of the frame | UDL ch. 12 | large, varied data sets |
| imitation learning | a policy from human driving | UDL ch. 19 | road vehicles |
| reinforcement learning | a policy from reward, by trial | UDL ch. 19 | simulators, legged robots |

### WHY THEY ARE LEFT OUT

Ex_11 follows one taped line with one camera, scored on rendered frames where the true position is known, so none of them is needed to do it. They matter for the full-size machines of the first section.

# Exercise 11

## ONE STUDY NOTEBOOK, FROM A TO B

- **1 – 3** — the virtual JetRacer, its frames, the detector
- **4** — a network that steers, trained deliberately
- **5 – 6** — obstacles, and the strategies to avoid them
- **7** — the loop rate, the GPU and fewer bits
- **8 – 10** — the report, and the mini projects



*the same task on the real car: a demonstration in the lab*

## THE QUESTION EACH PART ANSWERS

| | |
|---|---|
| **1 – 3** | when is a fixed rule enough? |
| **4** | what does deliberate data buy? |
| **5 – 6** | which strategy reaches B first? |
| **7** | how fast must the loop be? |

## REPORT QUESTION, TO CONCLUDE

Which avoidance strategy reaches B fastest without a collision, and what limits its speed: the errors of the networks, the loop rate, or the distance needed to react?

SUMMARY

# Key Takeaways

## VISION POLICY

| | |
|---|---|
| **what it is** | a CNN from a frame to an action |
| **function** | where the line is, a little ahead |
| **train, tune** | frames that show the mistakes |
| **vs a fixed rule** | no hand tuning, holds under glare |
| **used for** | steering cars, robots, vehicles |

## THE LOOP ON A GPU

| | |
|---|---|
| **what it is** | many cores, one operation each |
| **function** | all outputs of a convolution at once |
| **train, tune** | the whole loop, measured on the car |
| **vs accuracy** | judged together, rate first |
| **used for** | a short blind distance $d = v/f$ |

*Next: what blocks the path - obstacles, and what to do about them.*

# Questions

## 1 – 2 · A NETWORK THAT STEERS

**1** **Why can a convolutional network steer from a camera frame where a hand-tuned line detector fails?**

The detector follows fixed steps chosen by an engineer: it keeps the pixels brighter than the floor and steers to their centre, so anything bright is taken for the line. The network learns its own features from frames whose true position of the line is known. On clean frames the detector meets the agreed error; with a patch of glare its error rises to 0.08, while the network, shown glare in training, keeps it at 0.055. *UDL ch. 10; Ex_11 sections 3 – 4*

**2** **What must the training frames show for the network to steer a car that is drifting off the line?**

The network learns only what the frames show: the floor, the camera mount and the light it will drive in, the disturbances such as glare, and on purpose the positions the car reaches when it is already going wrong, labelled with the way back. Trained only along the line, the same network is five times worse beside it, which is exactly where a drifting car needs it. Same network, same number of frames: only the collection differs. *UDL ch. 8; Ex_11 section 4*

## 3 – 4 · THE LOOP ON A GPU

**3** **Why is a steering network judged by the rate of the control loop as well as by its error?**

Between two decisions the car drives on its last command with no new information, over a blind distance of its speed divided by the decisions per second. With the network steering the virtual car round the lap it stays within 6 cm up to 17 cm blind and swings 15 cm at 20 cm, so a slightly more accurate network that halves the loop rate is usually a worse car. The whole loop is timed, from capture to command, because the network is only one of its stages. *Ex_11 section 7*

**4** **Why does a convolutional network need a GPU to steer in time, and what must be measured on the car?**

Every output of a convolution is its own sum of products over a small window of the frame, independent of the others, so the many cores of a GPU compute them at the same time. A ResNet-18 needs about 1.8 billion multiply-adds for one 224 by 224 frame. The peak rate of the GPU is only a ceiling: the loop rate that sets the blind distance is measured on the car itself, from the frame being captured to the steering command. *UDL ch. 10; Ex_11 section 7*

NEXT

# L11.2 — Obstacle Avoidance

What blocks the path: a network that tells free from blocked, the strategies that steer around an obstacle, and the distance the car needs to react.

## BEFORE THE NEXT SESSION

| | |
|---|---|
| **read** | UDL ch. 10 |
| **run** | Ex_11's notebook, sections 1 to 4 |
| **bring** | your detector and network errors |
| **think** | what in your frames is an obstacle? |

*The real JetRacer drives the same task in a lab demonstration, with the throttle limited in software and one hand on the stop.*